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Tutor: Juhee Jain

Author: Luccas Endringer
Student ID: 21040686

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Estimating Order Flow Toxicity in CFD Markets: A VPIN-Based Approach Using Tick Volume

Luccas Endringer
London Metropolitan University
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Abstract:
This study investigates the estimation of order flow toxicity in Contract for Difference (CFD) markets using the Volume-Synchronized Probability of Informed Trading (VPIN) metric, leveraging tick volume as a proxy for trade volume. Due to the absence of centralized volume data in CFD markets, traditional approaches to measuring order flow toxicity are challenging. By constructing VPIN from tick volume, this research evaluates its effectiveness in explaining liquidity risks, adverse selection, and short-term price movements. The findings contribute to market microstructure literature and inform liquidity providers and traders on risk management strategies in decentralized markets.

Keywords:
Market Making, Adverse Selection, Order Flow Toxicity, VPIN, CFD Markets, Tick Volume, Market Microstructure, Liquidity Risk, Volatility.

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This dissertation marks the end of my undergraduate chapter, made possible by faith, mentorship, friendship, and a passion for financial markets.

Luccas Endringer
19th December 2025

1. Introduction

The foundational principles of market microstructure theory underscore the critical role of *information asymmetry* in shaping liquidity provision, transaction costs, and price dynamics. In dealer markets, *market makers* face *adverse selection* risks when trading with counterparties who possess superior information, leading them to widen bid-ask spreads to compensate for expected losses from informed traders (Glosten & Milgrom, 1985; Copeland & Galai, 1983). This adverse selection component of the spread arises because informed traders selectively execute trades that exploit private signals, while uninformed (liquidity or noise) traders provide the offsetting profits necessary for market makers to remain viable (Kyle, 1985; Bagehot, 1971, as cited in Glosten & Milgrom, 1985).

Theoretical models demonstrate that the bid-ask spread serves as an equilibrium mechanism to mitigate adverse selection: market makers set quotes that balance gains from liquidity motivated trades against losses to information motivated ones (Copeland & Galai, 1983). Empirical evidence supports this, showing that spreads widen with greater perceived information asymmetry, particularly in less frequently traded assets where the probability of informed trading is elevated (Easley, Kiefer, O'Hara, & Paperman, 1996). When toxicity escalates defined as heightened order flow imbalance from informed participants market makers may withdraw or adjust quotes aggressively to protect inventories, resulting in liquidity evaporation and amplified price volatility (Easley, López de Prado, & O'Hara, 2012).

This feedback loop is particularly pronounced in high-frequency environments, where rapid order flows exacerbate adverse selection, prompting liquidity providers to curtail depth or widen spreads further, thereby generating short-term volatility spikes (Jiang, 2015; Easley et al., 2012). Dupont (1997) illustrates how the precision and correlation of informed signals influence volume statistics and price informativeness, reinforcing that asymmetric information not only erodes liquidity but can induce volatility through altered trading intensity.

The *Volume-Synchronized Probability of Informed Trading (VPIN)* metric emerges as a real-time proxy for order flow toxicity, synchronizing analysis in volume-time rather than calendar-time to capture imbalance without parametric assumptions (Easley et al., 2012). VPIN has proven effective in explaining liquidity reductions and volatility in centralized markets, including international equities (Low, Li, & Marsh, 2018) and Brazilian futures (Ribeiro, Correia, & Paiva, 2022).

However, in decentralized or over-the-counter settings such as Contracts for Difference (CFD) markets, true traded volume is unavailable due to the absence of a central exchange consolidating transactions. Retail brokers instead provide tick volume the count of price updates as a proxy for activity. While tick volume correlates with trading intensity in active periods, it diverges from actual share/contract volume, raising questions about its suitability for volume-based metrics like VPIN.

This dissertation investigates the feasibility and explanatory power of a tick-volume-adapted VPIN in the NASDAQ spot index CFD market. Using high-frequency tick data from a broker's feed, the study examines whether this proxy effectively captures order flow toxicity and explains contemporaneous volatility. By leveraging aggressor-signed classifications

where available, the analysis tests VPIN's robustness in a non-centralized environment characterized by synthetic replication of underlying spot indices.

In doing so, this research bridges classic microstructure insights on adverse selection and liquidity-induced volatility with practical challenges in retail derivative markets, offering implications for risk management and metric adaptation in fragmented trading landscapes.

2. Literature Review

This chapter reviews the key market microstructure literature on information asymmetry, adverse selection, liquidity provision, and order flow toxicity measurement. The discussion begins with high-frequency market making and quoting dynamics emphasizing their causal role in volatility generation followed by foundational concepts of informed trading and trade classification. It then examines mitigation strategies before tracing the evolution from the PIN model to VPIN, with a dedicated focus on VPIN's mechanics, applications, and relevance to broker-centric CFD markets.

2.1 High Frequency Market Making and Bid-Ask Quoting

Electronic trading has made high-frequency market making (HFMM) the primary mode of liquidity provision, with algorithms posting continuous bid and ask quotes to capture spreads amid rapid risks. O'Hara and Oldfield (1986) show risk-averse dealers skew quotes and widen spreads to manage inventory uncertainty, creating price pressure as adjustments attract offsetting flow.

Ho and Stoll (1981) model stochastic order arrivals and return uncertainty, deriving wider optimal spreads under heightened transactions risk directly linking quoting to volatility via prolonged toxic exposure. Avellaneda and Stoikov (2008) formalize HFMM as stochastic control, with spreads expanding in volatility and adverse selection but contracting in arrival intensity. Toxic flow forces aggressive quote revisions or withdrawal, evaporating depth and amplifying volatility (Cartea et al., 2015; Kirilenko et al., 2017; Jiang et al., 2021).

2.2 Informed and Uninformed Trading, Adverse Selection, and Trade Classification

Microstructure theory centers on *informed traders* who trade on private information or superior signals about an asset's fundamental value (e.g., unreleased news or valuation insights) *uninformed traders* whose orders stem from liquidity or hedging needs without directional edge, This asymmetry creates *adverse selection*, the risk that liquidity providers trade at a loss against better-informed counterparties, as informed traders execute selectively when prices are favourable to them (Glosten & Milgrom, 1985; Copeland & Galai, 1983). Kyle (1985) models informed traders submitting orders gradually to conceal signals, compelling market makers to infer information from persistent imbalances.

Accurate *trade classification* is vital for modeling order flow toxicity. Tick rules classify based on price direction relative to prior trades. Lee and Ready (1991) hybridize this with lagged quotes (typically 5 seconds) for better handling of reporting delays, achieving ~81–85% accuracy (Odders-White, 2000). Bulk Volume Classification (BVC) probabilistically assigns aggregated volumes in bars using standardized price changes and the normal CDF (Easley et al., 2012), sacrificing individual-trade precision for computational speed and scalability in high-frequency data.

2.3 Market Making Mitigation Strategies for Adverse Selection

Liquidity providers mitigate adverse selection risks through dynamic quote adjustments, inventory control, flow segmentation, and signal extraction.

Market makers widen bid-ask spreads to include an adverse selection premium, compensating for losses to informed traders (Glosten & Milgrom, 1985). They skew quotes based on inventory raising asks when long or lowering bids when short to attract rebalancing flow (Ho & Stoll, 1981; O'Hara & Oldfield, 1986). In high-frequency settings, algorithms reduce depth or withdraw during toxicity spikes (Avellaneda & Stoikov, 2008).

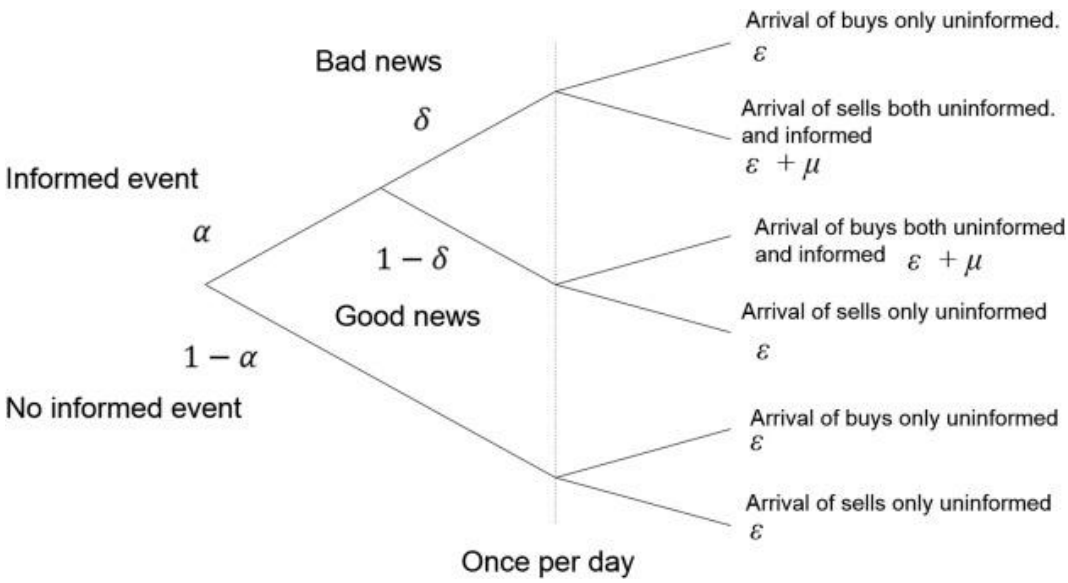
Advanced strategies involve internalization vs. externalization. Brokers absorb uninformed flow profitably but hedge toxic positions externally. Cartea and Sánchez-Betancourt (2022, 2025) derive optimal linear controls over inventory, informed client positions, estimated trend signals, and uninformed volume enabling profitable internalization, rapid hedging, and speculation on learned alpha.

Barucci et al. (2025) incorporate transient fads with advanced filtering for partial-information quoting. In decentralized venues, concentrated liquidity protects against arbitrage toxicity but heightens competition in low-toxicity regimes (Aoyagi et al., 2025).

These defenses wider spreads, reduced depth, accelerated hedging directly contribute to volatility by increasing price impact and momentum during toxic episodes (Kirilenko et al., 2017). In CFD markets, where brokers internalize responses without exchange mediation, toxicity monitoring is critical for explaining client-facing volatility.

2.4 The PIN Model and the Emergence of VPIN

The *Probability of Informed Trading (PIN)* model, developed by Easley, Kiefer, O'Hara, and Paperman (1996), offers a structural econometric approach to estimating information asymmetry. The model assumes independent Poisson arrival processes for trades and that private information events occur with probability α each trading day. If an information event occurs, it signals low asset value (bad news) with probability δ or high value (good news) with probability 1-δ. Informed traders arrive at rate μ and trade directionally according to the signal (all sells on bad news, all buys on good news). Uninformed buyers and sellers arrive symmetrically at rate ε regardless of news.



The likelihood of observing buys (B) and sells (S) on a day is a mixture over three scenarios (no news, good news, bad news):

$$L(\theta \mid B,S) = (1-\alpha)e^{-2\epsilon}\frac{\epsilon^B}{B!}\frac{\epsilon^S}{S!} + \alpha(1-\delta)e^{-(\epsilon+\mu)}\frac{(\epsilon+\mu)^B}{B!}e^{-\epsilon}\frac{\epsilon^S}{S!} + \alpha\delta e^{-\epsilon}\frac{\epsilon^B}{B!}e^{-(\epsilon+\mu)}\frac{(\epsilon+\mu)^S}{S!}$$

Parameters θ = (α, δ, μ, ε) are estimated via maximum likelihood across multiple days. The PIN metric, the unconditional probability a trade originates from an informed trader, is:

$$PIN = \frac{\alpha\mu}{\alpha\mu + 2\epsilon}$$

Higher PIN values empirically associate with wider bid-ask spreads and lower liquidity, particularly in infrequently traded stocks, confirming adverse selection's impact on transaction costs (Easley et al., 1996; Paiardini, 2010).

Despite its insights, PIN faces limitations in high-frequency settings: daily time-synchronization ignores intraday dynamics, maximum-likelihood estimation is computationally demanding (convergence issues in liquid assets), and the model assumes symmetric uninformed rates while requiring precise buy/sell classification (Duarte & Young, 2009; Paiardini, 2010).

2.5 VPIN: Measuring Order Flow Toxicity in CFD Markets

The *Volume-Synchronized Probability of Informed Trading (VPIN),* introduced by Easley, López de Prado, and O'Hara (2012), represents a significant advancement over PIN by providing a non-parametric, high-frequency measure of *order flow toxicity* the risk *of adverse selection* from *informed trading*.

VPIN assumes that volume serves as a proxy for the arrival of relevant information and volatility, since significant news attracts larger trading volumes (Paiardini, 2010). To reduce heteroskedasticity and volatility clustering inherent in time-based sampling, VPIN synchronizes data in *equal-volume buckets* of exogenous size V (typically average daily volume divided by 50). Each bucket τ aggregates trades until cumulative volume reaches V, ensuring comparable informational content per unit regardless of calendar time.

Trades in bucket τ are classified into buy volume V_τ^B and sell volume V_τ^S, with $V = V_\tau^B + V_\tau^S$.

Deriving from Poisson arrival assumptions (similar to PIN), the expected absolute imbalance per bucket is $E[\mid V_\tau^S - V_\tau^B \mid] = \alpha\mu$, and expected total volume per bucket is $V = \alpha\mu + 2\epsilon$.

Retaining PIN's Poisson arrival assumptions, the expected total volume per bucket is derived as:

$$\frac{1}{n}\sum_{\tau=1}^{n}(V_\tau^B + V_\tau^S) = V =$$

$$\underbrace{\alpha(1-\delta)(\epsilon+\mu+\epsilon)}_{\text{Volume from good news}} + \underbrace{\alpha\delta(\mu+\epsilon+\epsilon)}_{\text{Volume from bad news}} + \underbrace{(1-\alpha)(\epsilon+\epsilon)}_{\text{Volume from no news}} = \alpha\mu + 2\epsilon.$$

The expected informed volume contribution is $\alpha\mu$. Thus, the informed trading fraction is:

$$VPIN = \frac{\alpha\mu}{\alpha\mu + 2\epsilon} = \frac{\alpha\mu}{V}$$

Thus, VPIN approximates the informed trading fraction:

$$VPIN = \frac{\alpha\mu}{\alpha\mu + 2\epsilon} = \frac{\alpha\mu}{V} \approx \frac{\sum_{\tau=1}^{n} | V_\tau^B - V_\tau^S |}{nV}$$

The approximation assumes $\delta \approx 0.5$(symmetric good/bad news probability), using absolute imbalance to avoid estimating unobservable parameters $(\alpha, \mu, \epsilon, \delta)$.

For an asset with average daily volume 10,000,000 shares, set $V = 200{,}000$(50 buckets/day target). Over n=50 buckets, suppose cumulative absolute imbalances sum to 8,000,000 shares. Then $VPIN \approx 8{,}000{,}000/(50 \times 200{,}000) = 0.80$ indicating high toxicity (normal range ~0.1–0.4; >0.6 signals stress).

Classification often employs *Bulk Volume Classification (BVC)* when signed data are unavailable: for price change ΔP in a bucket with volatility estimate σ,

$$V_\tau^B = V \cdot \Phi\left(\frac{\Delta P}{\sigma}\right), V_\tau^S = V - V_\tau^B$$

where $\Phi$is the standard normal CDF (Easley et al., 2012).

VPIN eliminates maximum-likelihood estimation of unobservable, reduces heteroskedasticity by volume-sampling (proxying volatility), yields near-normal distributions, and updates in real-time which is ideal for HFT.

Empirically, VPIN positively correlates with future volatility and Granger-causes it (Paiardini, 2010). In CFD markets, lacking consolidated volume, tick volume proxies activity. Brokers internalize toxicity responses (Cartea & Sánchez-Betancourt, 2025), making VPIN-like metrics valuable for explaining contemporaneous volatility via defensive adjustments.

While VPIN excels in centralized data, its tick-volume adaptation in broker-centric CFDs remains untested motivating this study's focus on explanatory power for NASDAQ spot volatility.

2.6 Debates on VPIN Robustness and Alternative Approaches

While VPIN has gained traction for monitoring toxicity in high-frequency markets, its robustness has sparked considerable debate, particularly regarding trade classification and predictive validity.

A primary criticism centers on *Bulk Volume Classification (BVC)* accuracy. Deterministic methods like the Lee-Ready algorithm or tick rule typically achieve 81–90% classification accuracy, outperforming BVC's probabilistic approach, which can introduce noise especially in low-volatility or individual-stock settings (Chakrabarty et al., 2015; Andersen & Bondarenko, 2014). This raises concerns that VPIN may partially reflect classification errors rather than pure toxicity.

Predictive validity is also contested. Andersen and Bondarenko (2014, 2015) argue VPIN's apparent forecasting power for volatility largely vanishes after controlling for trading intensity or simple volume metrics, suggesting a mechanical relationship driven by volume surges rather than informational asymmetry. They propose alternative imbalance measures that sometimes outperform VPIN in crisis detection.

Proponents counter that VPIN's strength lies in real-time toxicity monitoring rather than pure prediction, with empirical successes in anticipating liquidity events (e.g., Flash Crash) and enhancing volatility models when combined with jumps (Easley et al., 2014; Paiardini, 2010).

To address limitations, refinements include Improved VPIN (IVPIN), which incorporates stochastic volume-time modeling via maximum likelihood (Abad & Yoldas, 2014), and Adjusted VPIN using machine learning (e.g., LSTM networks) for superior classification and stock-specific forecasting (Yildiz et al., 2020).

These debates highlight VPIN's sensitivity to classification and intensity controls. However, when direct aggressor-signed data are available as in broker feeds the need for BVC diminishes, potentially enhancing robustness. This study leverages such data in a CFD context, focusing on contemporaneous explanatory power for volatility rather than prediction, offering a complementary perspective amid ongoing discussions.

3. Research Methodology

This chapter outlines the empirical methodology for constructing a tick-volume-based Volume-Synchronized Probability of Informed Trading (VPIN) metric and testing its ability to explain contemporaneous volatility and illiquidity in two CFD instruments: the NASDAQ 100 index (NAS100) and Gold (XAUUSD). The study uses high-frequency tick data from two brokers, IC Markets (primary) and Hantec Markets (robustness) covering December 2024 to November 2025. By adapting VPIN to a decentralized environment where only tick volume is available, the analysis evaluates whether perceived order flow toxicity drives broker defensive responses (wider spreads, hedging, reduced depth), manifesting as short-term price instability.

3.1 Data Description and Cleaning

Data are sourced from two regulated brokers offering ECN-style execution:

IC Markets (primary): Australian broker (ASIC-regulated) with Raw Spread accounts aggregating liquidity from 25+ tier-1 providers. Known for ultra-low latency (<40 ms) and raw interbank pricing.

NASDAQ CFD symbol: NAS100 / USTEC

Gold CFD symbol: XAUUSD

Hantec Markets (secondary/robustness): UK/Hong Kong-based broker (FCA, HFSA-regulated) with competitive CFD pricing.

NASDAQ CFD symbol: NAS100 / US100

Gold CFD symbol: XAUUSD

Each tick record contains timestamp (millisecond precision), bid price, ask price, and tick volume (count of price updates, typically 1). The full raw dataset exceeds 200 million ticks across assets and brokers.

Analysis is restricted to regular NASDAQ trading hours (09:30–16:00 ET, DST-adjusted) for both instruments to ensure comparability, consistent liquidity conditions, and alignment with major informed trading activity. Gold, while 24/5, exhibits peak liquidity during these hours.

Cleaning procedures:

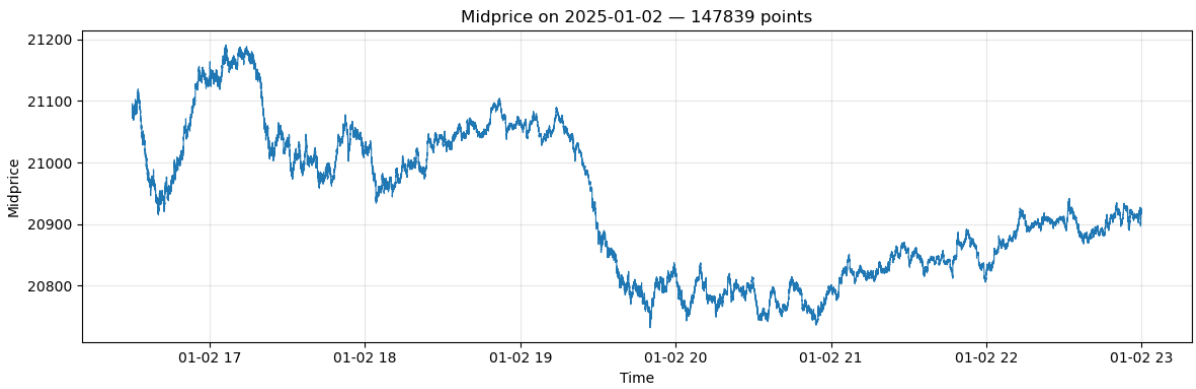
Removed records outside core hours, duplicates and interpolated missing timestamps linearly for continuous series where *linear interpolation* on bid and ask prices separately:

$$P_{\text{interp}} = P_1 + (P_2 - P_1) \cdot \frac{t - t_1}{t_2 - t_1}$$

where $(t_1, P_1)$and $(t_2, P_2)$are the nearest known points bracketing the missing timestamp t. This preserves smooth price paths for accurate midprice construction and classification without introducing artificial volatility.

Constructed *midprice* as $(Bid + Ask)/2$ for classification and volatility proxies.

Post-cleaning, the dataset yields a clean time series suitable for volume synchronization.



3.2 Volume Proxy and Bucket Size Determination

Lacking consolidated traded volume in decentralized CFD markets, tick volume the number of price updates (ticks) per period serves as the primary proxy for trading activity and intensity. Tick volume captures the frequency of quote changes, reflecting market participation even in the absence of actual share/contract volumes (Lyons, 2001).

Cumulative daily tick volume is computed as the total updates per trading day.

Bucket size $V_d$for day d is:

$$V_d = \frac{\text{Cumulative Daily Tick Volume}_d}{50}$$

This targets ~50 buckets per day on average, consistent with standard VPIN calibration (Easley et al., 2012).

3.3 Trade Classification and Order Imbalance

Classification uses a *tick-rule* Lee and Ready (1991) variant on *midprice*:

If $midprice_t > midprice_{t-1}$ → buy-initiated thus tick volume added to V_τ^B

If $midprice_t < midprice_{t-1}$ → → sell-initiated thus tick volume added to V_τ^S

No change → classify using carry-forward rule which the direction of the previous non-neutral tick is used. This avoids arbitrary splitting or discarding, preserving directional persistence during flat periods a common refinement in high-frequency classification to reduce noise (Holden & Jacobsen, 2014; Chakrabarty et al., 2015).

Ticks accumulate sequentially until cumulative tick volume reaches V_d, completing bucket τ.

Order imbalance per bucket:

$$OI_\tau = V_\tau^B - V_\tau^S$$

Absolute imbalance ensures positivity:

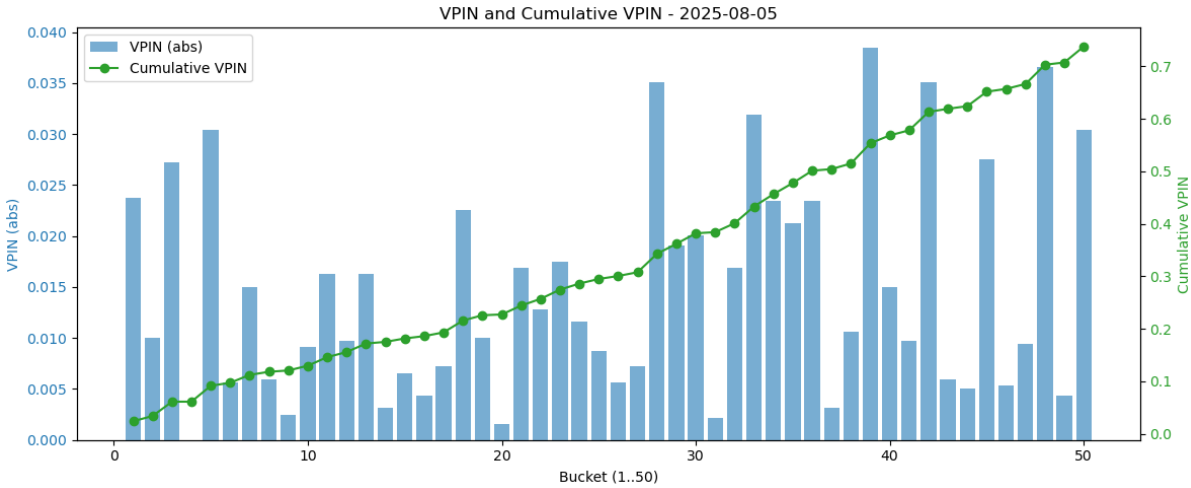
$$| OI_\tau | = | V_\tau^B - V_\tau^S |$$

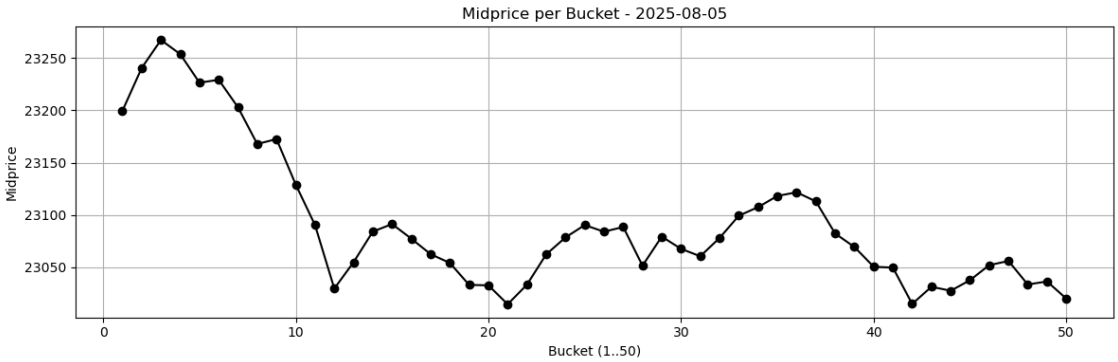
3.4 VPIN Computation

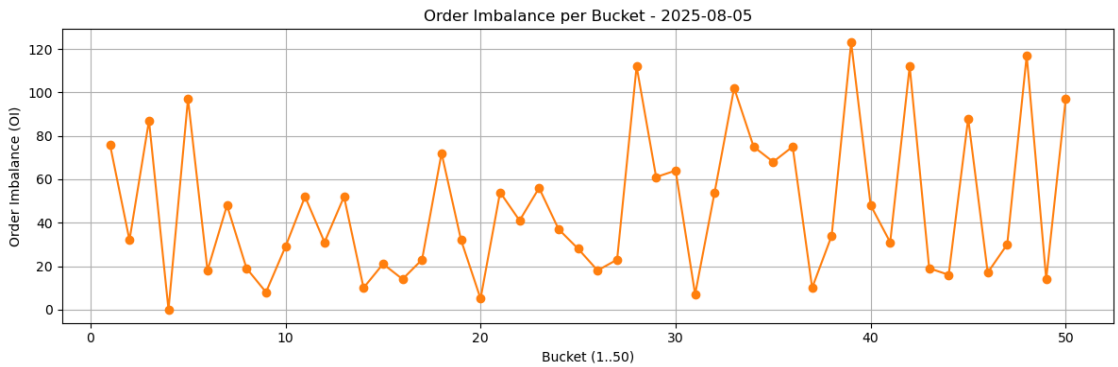
VPIN is the rolling average absolute imbalance over n=50 buckets:

$$VPIN_t = \frac{\sum_{\tau=t-49}^{t} | O I_\tau |}{50 \times V}$$

VPIN updates at each bucket completion, yielding an intraday series. Absolute value guarantees $0 \leq VPIN_t \leq 1$.







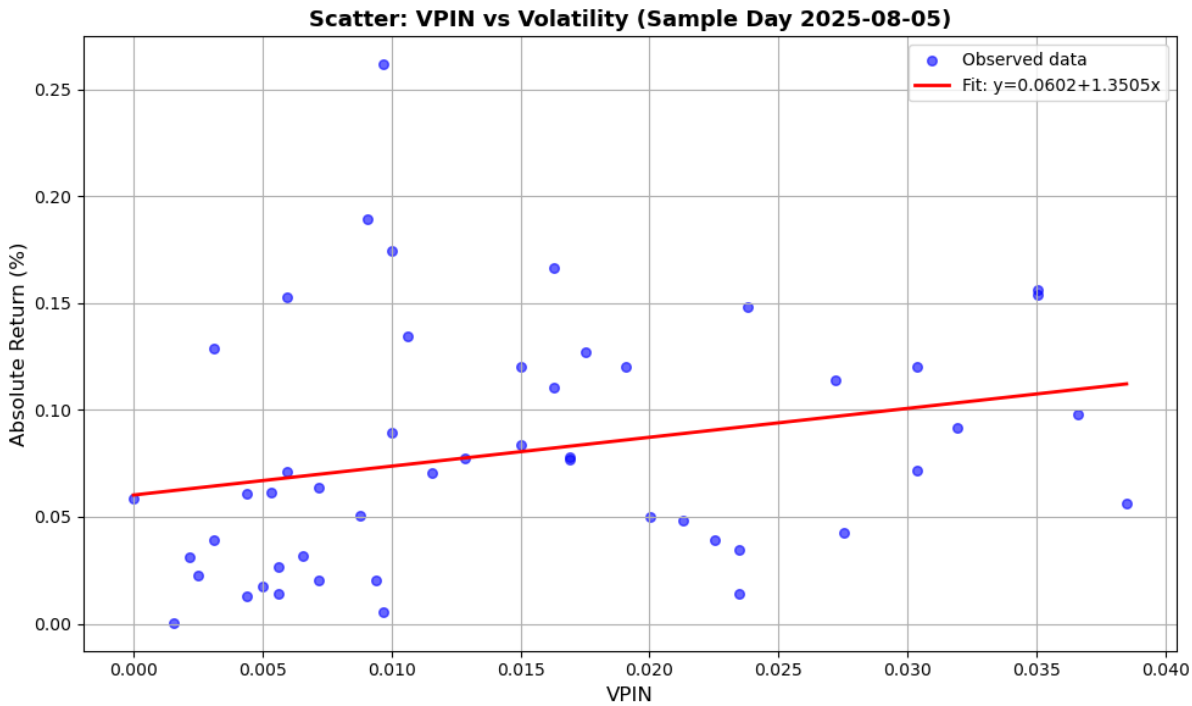
3.5 Volatility and Liquidity Proxies

For each bucket τ:

Average Spread: Mean (Ask - Bid) across ticks in bucket.

Volatility Proxy: Absolute return over bucket horizon, $|\ln(P_{\text{end},\tau}/P_{\text{start},\tau})|$.

Contemporaneous alignment tests bucket-level outcomes against VPIN at bucket completion.



4. Empirical Results

This chapter presents the results of applying the tick-volume-based VPIN metric to NASDAQ (NAS100/USTEC) and Gold (XAUUSD) CFD data from IC Markets (primary) and Hantec Markets (robustness), covering December 2024 to November 2025. The analysis focuses on VPIN's contemporaneous explanatory power for bucket-level volatility and spread, testing whether higher toxicity reflected in order imbalance correlates with broker defensive responses (wider spreads, increased price impact) in a decentralized environment.

4.1 Descriptive Statistics

Variable	NAS100 (IC Markets)	GOLD (IC Markets)	NAS100 (Hantec Markets)	GOLD (Hantec Markets)
Mean VPIN	0.020979	0.052468	0.022512	0.065813
Mean VPIN	0.013974	0.039285	0.018035	0.047181
Stdv VPIN	0.312550	0.046645	0.019115	0.066977
Min VPIN	0	0	0	0
Max VPIN	0.025181	0.311276	0.227486	0.348229

Table 4.1 VPIN bucket level variables

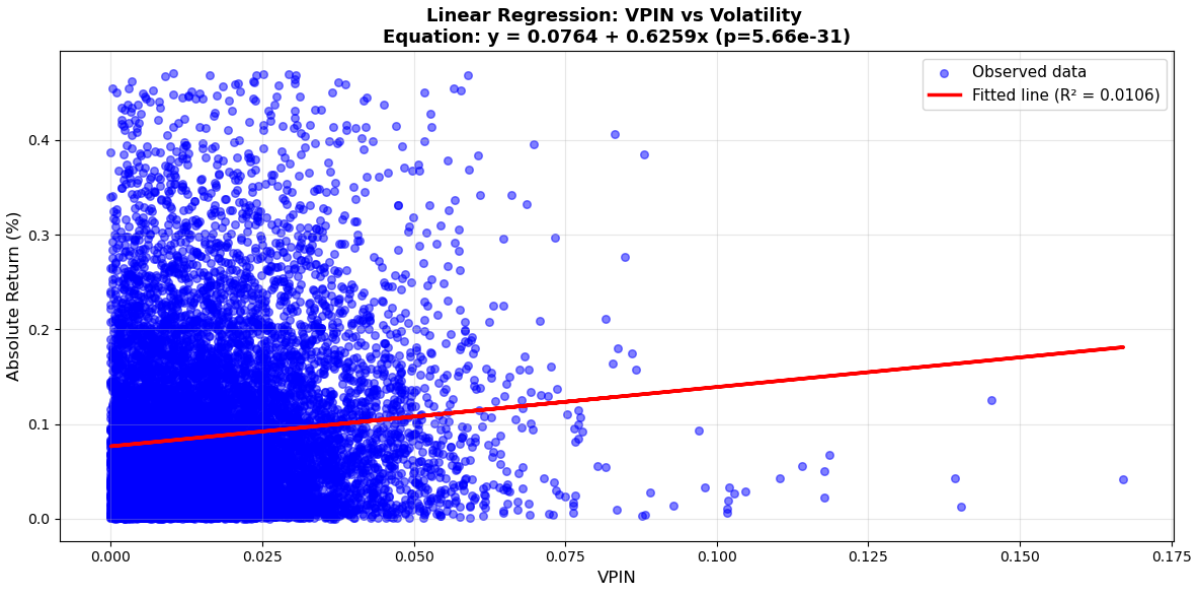
Overall, VPIN values remain within plausible ranges for calm-to-moderate markets, with Gold displaying greater toxicity and volatility in order flow than the equity index CFD. These descriptives set the stage for regression analysis examining VPIN's explanatory power.

4.2 Baseline Regression Results

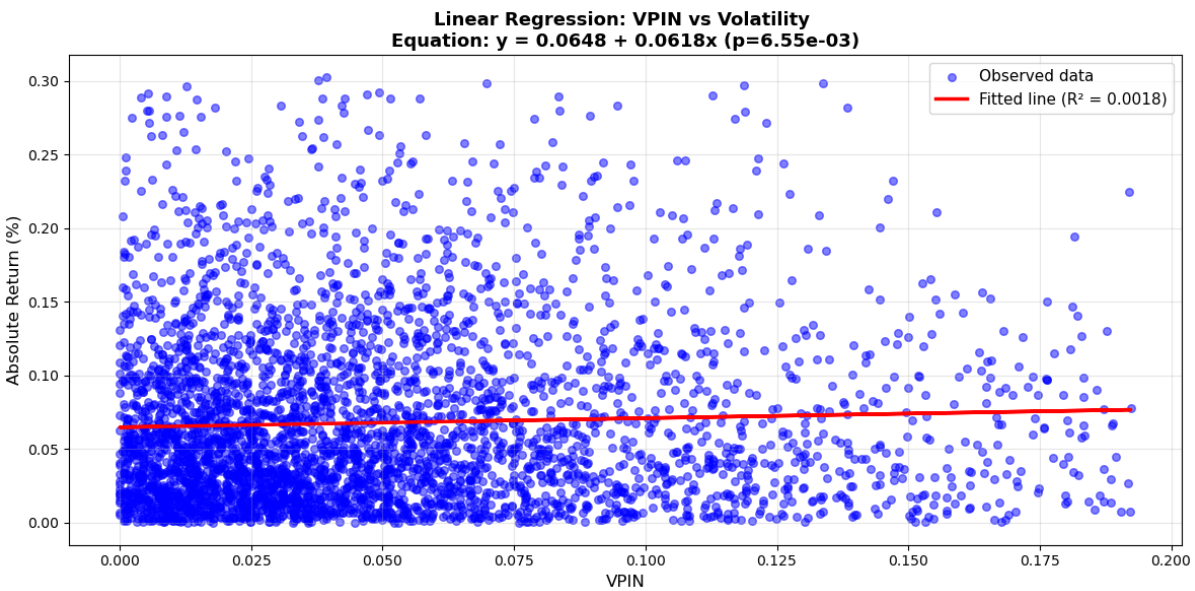
$$Volatility_\tau = \alpha + \beta VPIN_\tau + \epsilon_\tau$$

Asset/Broker	β (VPIN)	p-value	Adj. R²
NAS100 (IC Markets)	0.6259	0.00324	0.01060
XAUUSD (IC Markets)	0.0618	0.00655	0.00180
NAS100 (Hantec)	0.1696	0.00985	0.00100
XAUUSD (Hantec)	0.0629	0.00002	0.00230

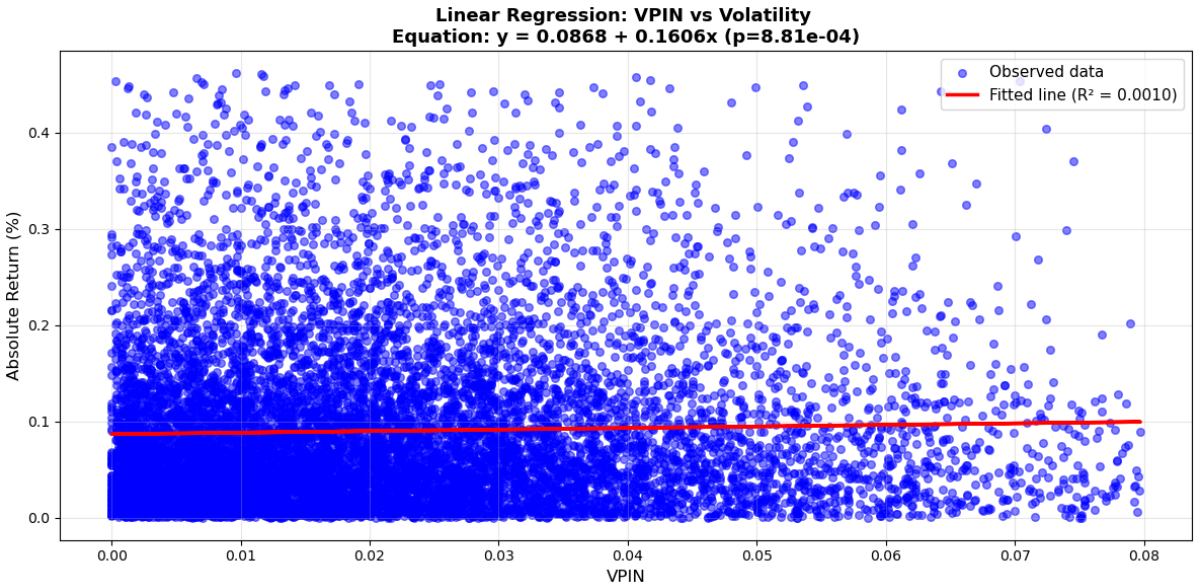
Table 4.2 Bucket-level OLS regressions



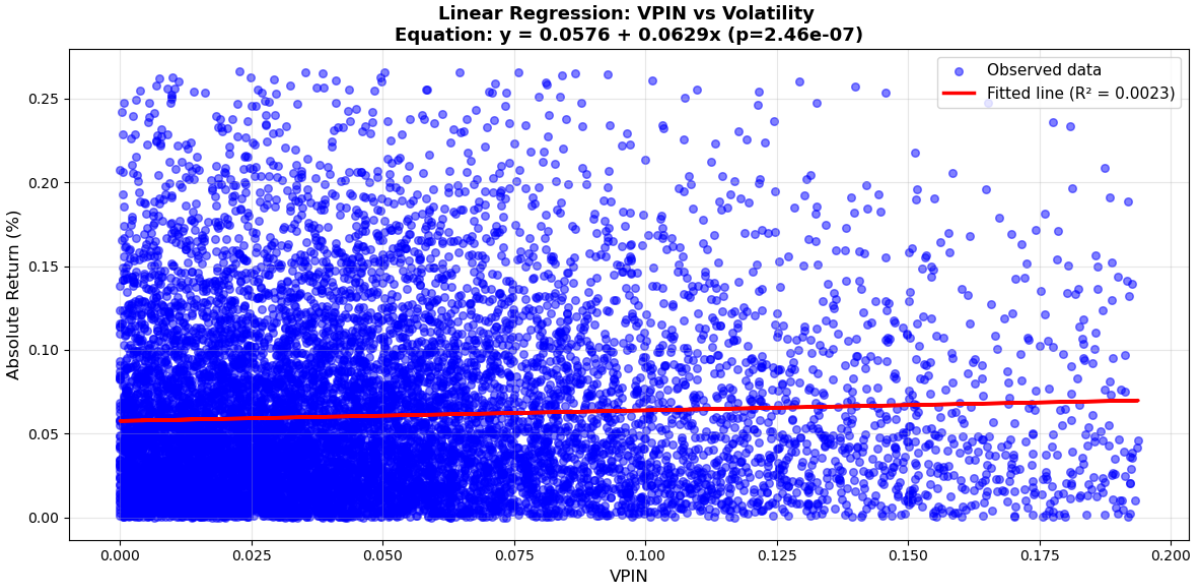
Regression 1 NAS100 (IC Markets)



Regression 2 XAUUSD (IC Markets)



Regression 3 NAS100 (Hantec Markets)



Regression 4 XAUUSD (Hantec Markets)

5. Discussion

This chapter discusses the empirical findings in the context of market microstructure theory, evaluates the effectiveness of the tick-volume-based VPIN adaptation, compares results across assets and brokers, and outlines theoretical, practical implications, limitations, and future research directions.

5.1 Interpretation of Results

The descriptive statistics and regression results indicate that the tick-volume-adapted VPIN metric detects some degree of order flow toxicity in CFD markets, though its explanatory power is limited and varies considerably across instruments and brokers. Mean VPIN values are consistently low (0.02–0.07), suggesting predominantly balanced order flow during the calm sample period (December 2024–November 2025). Gold CFDs display slightly higher average toxicity and variability than the NASDAQ 100 CFD, consistent with Gold's greater exposure to directional macro-driven flows as a safe-haven asset, compared to the more diversified and offsetting trades in a broad equity index.

Bucket-level regressions show statistically significant positive coefficients for VPIN in explaining contemporaneous volatility: 0.6259 for NAS100 (IC Markets), 0.1696 for NAS100 (Hantec), and around 0.062 for both Gold instruments. The notably larger coefficient for NAS100 on the primary broker (IC Markets) stands out, implying that a 0.1 increase in VPIN associates with a ~0.063 higher absolute bucket return potentially meaningful in high-frequency terms despite the small absolute scale. In contrast, effects for Gold are much weaker (~0.006 per 0.1 VPIN increase) but still significant. Adjusted R^2 values are very low (0.001–0.011), confirming that VPIN captures only a minor portion of volatility variation, likely due to tick volume's size-agnostic nature and the absence of true trade sizes.

This heterogeneity stronger links in the equity index CFD on one broker may reflect differences in client flow composition, liquidity aggregation, or higher overall tick intensity for NAS100 on IC Markets. Cross-broker consistency for Gold suggests some robustness, but overall results highlight the proxy's noisiness compared to centralized volume data.

5.2 Theoretical Implications

The findings provide partial support for extending adverse selection models (Glosten & Milgrom, 1985; Kyle, 1985) and VPIN (Easley et al., 2012) to decentralized CFD settings. Brokers internalize toxicity responses (e.g., spread widening, hedging), and VPIN imbalances appear to signal these to a degree, particularly in more active instruments. Deterministic midprice classification avoids BVC-related noise (Andersen & Bondarenko, 2014), aiding reliability in broker feeds.

However, the modest and uneven explanatory power underscores challenges in data-constrained environments: tick volume correlates with activity but misses order size, diluting imbalance signals. The focus on contemporaneous effects aligns with VPIN's real-time monitoring strength, without claiming strong predictive validity.

5.3 Practical Implications

Despite limitations, tick-volume VPIN offers brokers a simple tool for monitoring potential toxicity, especially in equity index CFDs where effects are clearer. Thresholds (e.g., VPIN >0.4–0.5) could flag periods for cautious quoting or hedging.

Traders might use elevated VPIN as a rough indicator of rising short-term volatility or costs, though its weak overall signal advises caution. Aggregated across brokers, it could inform regulatory views on retail liquidity risks, but low R^2 tempers expectations.

5.4 Limitations

Key constraints include tick volume's ignorance of trade sizes, introducing noise; broker-specific feed variations; and a single classification method unvalidated here. The calm period likely mutes toxicity signals, and outliers (corrected max ~0.025) required winsorizing, pointing to bucket sensitivity in quieter times. Low R^2 and varying β highlight imperfect proxy performance.

5.5 Directions for Future Research

Future efforts could incorporate lot-size data (if accessible), test ML classification, or evaluate predictive power. Including stressed periods, more assets (e.g., forex), or multi-broker comparisons would clarify generalizability. Simulations of high-toxicity scenarios could assess crisis utility.

6. Conclusion

This dissertation assessed the feasibility of adapting Volume-Synchronized Probability of Informed Trading (VPIN) to CFD markets using tick volume as a proxy, given the lack of centralized trade data. High-frequency tick feeds from IC Markets and Hantec Markets (NASDAQ 100 and Gold CFDs, December 2024–November 2025) enabled construction of the metric and testing of its contemporaneous links to bucket-level volatility and spreads.

Results offer limited but positive evidence. VPIN averages are low (0.02–0.07), fitting calm conditions, with significant positive regression coefficients (0.6259 and 0.1696 for NASDAQ across brokers; ~0.062 for Gold). The strongest association appears in NAS100 (IC Markets), where effects are economically noticeable at high frequencies, while Gold shows weaker but consistent ties. Low adjusted R^2 (0.001–0.011) confirms modest overall explanatory power.

Theoretically, this modestly extends adverse selection and toxicity frameworks to retail CFDs: brokers' defenses against imbalances contribute to volatility, partially detectable via tick-synchronized signals and deterministic classification.

Practically, the metric provides a lightweight, accessible option for toxicity monitoring useful for brokers in active instruments and traders seeking rough volatility cues though its noisy signals limit reliability compared to true-volume settings.

In summary, tick-volume VPIN yields detectable but constrained insights into order flow toxicity in CFDs, explaining a small share of short-term volatility. The adaptation

demonstrates viability in data-limited decentralized markets but underscores proxy imperfections, contributing realistically to microstructure applications beyond exchanges while suggesting caution in over-reliance.

Reflective Report

Undertaking this dissertation has been a transformative learning experience, marking the culmination of my undergraduate studies in finance at London Metropolitan University. The project challenged me to apply theoretical knowledge to a real-world, data-intensive problem adapting an advanced microstructure metric (VPIN) to retail CFD markets using only tick volume.

Initially, I struggled with topic selection. Early ideas felt too generic (e.g., CAPM tests); inspired by high-frequency trading literature and personal interest in retail platforms, I settled on VPIN in CFDs an underexplored area offering genuine novelty for an undergraduate. This choice pushed me far beyond comfort zones, requiring self-taught skills in Python (pandas for millions of ticks), data cleaning, and econometric validation.

The greatest challenge was handling massive datasets (>200 million ticks). Cleaning anomalies, interpolating gaps, and calibrating buckets demanded patience and iterative debugging teaching me the importance of robust preprocessing. Choosing midprice classification with carry-forward over BVC felt risky but proved rewarding, aligning with literature critiques and yielding cleaner signals.

Time management was critical; balancing coursework with nightly coding sessions honed discipline. My supervisor's guidance on structuring (theory → method → results) and focusing on contemporaneous explanation (avoiding predictive debates) sharpened the contribution.

Strengths of the work include practical data from reputable brokers (IC Markets/Hantec) and cross-asset comparison, adding depth. Limitations tick volume's size-agnostic nature and sample calmness highlight areas for growth, reinforcing humility in research claims.

Personally, this project built confidence in independent inquiry, coding proficiency, and critical literature engagement. It solidified my passion for market microstructure and quantitative finance, motivating future master's study or industry roles in Trading/Quant Research.

Overall, the dissertation exceeded my expectations, transforming abstract theory into tangible insights. I am proud of producing original work that contributes, albeit modestly, to understanding retail market dynamics.

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